

PAMPA, TX, USA

WMO: 720313

Lat: 35.613N

Lon: 100.996W

Elev: 3245

StdP: 13.05

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03052

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.9	14.9	-4.0	5.0	15.9	1.9	6.8	25.9	33.0	48.7	30.1	49.9	10.1	10	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	99.2	66.3	96.7	66.7	94.0	66.7	72.7	86.9	71.3	85.7	70.2	84.3	13.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.1	120.7	77.5	67.8	115.2	76.5	66.2	109.0	75.6	38.7	87.5	37.3	85.8	36.2	83.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.7	26.8	24.5	1.4	103.7	4.2	3.4	-1.6	106.2	-4.1	108.1	-6.4	110.0	-9.5	112.5		
(5)			0.5	73.9	3.9	2.2	-2.3	75.5	-4.6	76.8	-6.8	78.1	-9.7	79.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	37.7	40.3	50.0	57.2	65.8	76.3	79.3	77.9	71.1	58.6	47.9	38.1		
	DBStd	17.43	10.47	11.75	10.20	9.50	9.05	5.63	5.09	5.55	6.88	9.54	9.89	10.17		
	HDD50	1443	398	306	126	39	6	0	0	0	0	33	153	382		
	HDD65	3992	846	692	472	264	101	2	1	1	24	240	516	833		
	CDD50	4524	17	36	126	254	497	788	908	864	632	299	89	14		
	CDD65	1596	0	1	6	30	127	339	444	400	206	41	2	0		
	CDH74	18766	5	40	175	561	1591	3925	5374	4339	2144	533	77	2		
	CDH80	9278	0	11	41	203	694	1979	2951	2282	934	173	10	0		
	WSAvg	12.2	11.5	12.4	13.4	15.0	13.1	13.5	11.1	10.2	11.4	11.9	12.1	11.5		
	PrecAvg	23.1	0.8	0.6	1.3	2.2	3.4	2.9	2.7	3.1	1.7	2.2	1.0	1.1		
(15) Precipitation	PrecMax	34.0	2.0	2.3	5.8	11.4	6.1	9.5	7.1	5.5	4.3	8.0	3.7	4.2		
	PrecMin	16.7	0.0	0.0	0.0	0.0	0.3	0.0	0.4	0.3	0.0	0.2	0.0	0.1		
	PrecStd	3.9	0.6	0.6	1.3	2.4	1.5	2.1	1.6	1.4	1.3	1.9	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.6	79.1	84.2	90.8	96.5	102.5	103.6	100.7	96.5	90.6	81.4	70.5		
		MCWB	49.0	50.6	55.0	56.8	59.7	65.6	65.4	67.5	65.0	62.6	53.3	47.9		
	2%	DB	67.8	71.9	79.2	86.1	91.4	97.5	99.0	98.6	93.0	84.5	75.3	64.4		
		MCWB	47.5	48.9	53.5	55.7	61.1	66.0	66.5	67.5	65.4	61.7	52.6	46.5		
	5%	DB	60.9	66.1	73.1	81.2	88.1	94.4	96.5	95.3	89.9	80.5	70.1	59.7		
		MCWB	44.2	46.8	51.5	55.2	61.2	66.7	67.3	67.9	65.5	59.4	50.9	44.6		
	10%	DB	54.8	59.8	68.2	75.1	83.1	90.4	93.3	91.3	86.0	75.1	64.1	54.5		
		MCWB	41.1	43.8	49.7	54.3	61.1	66.8	67.7	67.2	64.6	58.1	48.7	42.2		
	0.4%	WB	50.9	54.4	60.6	64.1	71.1	75.0	74.1	74.6	71.4	69.1	59.2	53.0		
		MCDB	69.8	71.8	71.7	75.0	82.2	90.9	87.2	90.3	84.7	77.7	67.6	61.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.1	50.4	57.5	61.3	68.4	72.2	72.3	72.5	69.6	66.0	55.4	48.5		
		MCDB	64.3	67.8	69.4	72.8	78.5	86.6	86.7	87.0	81.7	76.3	66.2	61.5		
	5%	WB	45.1	47.9	55.0	58.8	66.3	70.7	70.9	71.0	68.4	63.7	53.0	45.7		
		MCDB	60.1	64.7	66.6	70.5	78.3	85.2	86.2	84.9	80.8	72.9	66.1	57.8		
	10%	WB	41.9	45.1	52.1	56.5	64.2	69.3	69.5	69.8	67.2	61.0	50.6	43.0		
		MCDB	53.6	59.5	65.3	70.1	76.2	83.4	85.0	83.3	79.7	70.8	63.5	54.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.9	26.3	27.2	27.0	24.9	24.9	24.6	24.4	24.4	25.2	25.9	23.5		
		MCDBR	33.3	35.1	33.6	34.6	30.9	29.8	28.4	28.6	28.7	30.4	32.2	31.2		
		MCWBR	17.7	16.9	14.9	14.2	11.7	9.2	7.3	7.3	8.5	12.4	15.2	16.8		
	5% WB	MCDBR	31.9	33.3	28.5	25.9	25.4	25.1	24.5	23.9	22.5	23.5	27.7	29.2		
		MCWBR	17.4	16.7	14.1	13.3	12.0	10.3	8.1	7.8	8.1	11.5	14.3	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.264	0.294	0.326	0.342	0.378	0.381	0.371	0.339	0.303	0.276	0.257		
	taud		2.592	2.587	2.502	2.425	2.444	2.382	2.412	2.404	2.475	2.564	2.567	2.596		
	Ebn at Noon		307	316	311	304	298	286	284	286	292	296	296	300		
	Edh at Noon		23	27	32	36	36	38	37	36	32	27	24	22		
(44) All-Sky Solar Radiation	RadAvg		967	1216	1600	1978	2191	2345	2284	2040	1742	1312	1021	835		
	RadStd		85	118	114	123	182	156	100	106	86	156	104	61		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	N/A	(47)

Nomenclature: See separate page